

Mamba-370m chain-of-thought — success & failure per CoT task

HAR-CoT · ✓ SUCCESS

pred: sitting | gold: sitting

The accelerometer data over the 2.56 s window shows relatively low variability and consistent patterns across the X, Y, Z axes. The lack of significant peaks or troughs indicates no rapid or intense motion — characteristic of a stationary, low-intensity activity where the body stays mostly fixed with minor adjustments. → Answer: sitting.

HAR-CoT · ✗ FAILURE

pred: sitting | gold: lying

Stable readings across all three axes after an initial transient. The low variability and stability indicate a lack of intense or repetitive motion, aligning with a stationary, low-intensity activity. Therefore the activity is sitting. → Answer: sitting.

*Why: sitting & lying are both stationary, low-variability signals — the reasoning is sound but accelerometer magnitude can't separate the two postures (lying had the lowest HAR per-class F1).*

Sleep-CoT · ✓ SUCCESS

pred: Wake | gold: Wake

The signal shows frequent fluctuations with noticeable amplitude variability, suggesting heightened brain activity. The irregularity and complexity are associated with a more active brain state — consistent with an alert, conscious condition rather than deep sleep. → Answer: Wake

Sleep-CoT · ✗ FAILURE

pred: Non-REM stage 2 | gold: Non-REM stage 1

Rhythmic oscillations with moderate amplitude and frequency, associated with reduced consciousness and not as high-amplitude as alert states. The regularity and stability of the brain's electrical activity fit a sleep stage of reduced consciousness. → Answer: Non-REM stage 2.

*Why: N1 & N2 are adjacent NREM stages with overlapping EEG — correct 'reduced consciousness' read, but landed one stage off (the classic sleep-scoring error).*